ON L^p DENSITY OF SELF SIMILAR MEASURES AND BERNOULLI CONVOLUTIONS

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Abstract. We prove that an absolutely continuous self-similar measure has an L^p density for some $p > 1$. Moreover, explicit examples of Bernoulli convolutions with a continuous density are established.

1. Introduction

Self-similar measures are central objects in fractal geometry [Fal14, BSS23]. Understanding when they are absolutely continuous and establishing results about their regularity is intricate. In this paper, we establish the fundamental property that an absolutely continuous self-similar measure has a density in L^p for some $p > 1$. Moreover, we apply the quantitative proof method of the latter result to establish explicit examples of Bernoulli convolutions with a continuous density going beyond the only previously known case of Garsia numbers. All new results of this paper are formalised in Lean.

A contracting similarity on $\mathbb{R}^d$ is a map of the form $x \mapsto \rho U x + b$ with $\rho \in (0,1)$, U a $d \times d$ orthogonal matrix and $b \in \mathbb{R}^d$. Let $\mu = \sum_{i=1}^k p_i \delta_{g_i}$ be a probability measure supported on finitely many contracting similarities on $\mathbb{R}^d$. The self-similar measure of μ is the unique probability measure ν on $\mathbb{R}^d$ satisfying

$$\mu * \nu := \sum_{i=1}^{k} p_i (g_i)_* \nu = \nu. \tag{1.1}$$

Ever since Hutchinson's 1981 paper [Hut81], where it is proved that they exist and have compact support, self-similar measures have been well studied. We establish the following observation about self-similar measures.

Theorem 1.1. *Let μ be a probability measure supported on finitely many contracting similarities of $\mathbb{R}^d$ and denote by ν the self-similar measure of μ. If ν is absolutely continuous with respect to the Lebesgue measure, then the density of ν is in $L^p(\mathbb{R}^d)$ for some $p > 1$.*

The proof of Theorem 1.1 is short and combines self-similarity with a general higher integrability result. The key lemma of the proof and this paper is Lemma 2.1, which is a version of the reverse Hölder criterion of Spadaro [Spa12], as further developed by Kinnunen–Kurki–Mudarra [KKM22]. Lemma 2.1 assumes for an absolutely continuous measure ν on $\mathbb{R}^d$ control over the mass of $\nu(E)$ for sets E of small measure and then establishes a regularity gain for ν. The proof of Theorem 1.1 will then follow by showing that self-similarity prevents the mass of $\nu(E)$ being too large. Similar higher integrability arguments are used by Giaquinta–Modica [GM86, Theorems 3.1 and 4.1] in the regularity theory of elliptic equations and by De Lellis–Spadaro [DLS14, Section 5] in geometric measure theory.

The exponent $p > 1$ in Theorem 1.1 and Lemma 2.1 can be estimated quantitatively from the mass that ν assigns to sets of small Lebesgue measure. This insight allows us to use recent methods for establishing explicit examples of absolutely continuous self-similar measures [Var19, Kit24, KK25] to obtain suitable measure estimates, yielding quantitative L^p results that we use to construct Bernoulli convolutions with continuous densities.

The Bernoulli convolution ν_λ of parameter $\lambda \in (0,1)$ is the self-similar measure of the balanced probability measure on the similarities $x \mapsto \lambda x + 1$ and $x \mapsto \lambda x - 1$ with $x \in \mathbb{R}$. Another way to describe them is as the law of the random variable $\sum_{n=0}^{\infty} \xi_n \lambda^n$, where the ξ_n are independent and satisfy $\mathbb{P}(\xi_n = 1) = \mathbb{P}(\xi_n = -1) = 1/2$ for all $n \geq 0$. The study of Bernoulli convolutions has a rich history and we refer to surveys by Peres–Schlag–Solomyak [PSS00] and Varjú [Var18].

We highlight that Solomyak [Sol95] showed in 1995 that ν_λ is in L^2 for almost all $\lambda \in (1/2, 1)$. Another important result is due to Shmerkin [Shm19, Theorem 1.3], who showed that ν_λ has a density in L^p for every finite $p > 1$ and all parameters $\lambda \in (1/2, 1)$ outside a set of Hausdorff dimension zero.

Splitting the series $\sum_{n\geq 0} \xi_n \lambda^n$ into even and odd n gives $\nu_\lambda = \nu_{\lambda^2} * (D_\lambda)_* \nu_{\lambda^2}$ with $D_\lambda(x) = \lambda x$. Therefore, by Shmerkin [Shm19, Theorem 1.3] applied to L^2, for all $\lambda \in (1/\sqrt{2}, 1)$ except for a set of Hausdorff dimension zero, the factors ν_{λ^2} and $(D_\lambda)_* \nu_{\lambda^2}$ are both in L^2. So it follows that ν_λ has a continuous density for $\lambda \in (1/\sqrt{2}, 1)$ outside a set of Hausdorff dimension zero. By a similar argument using Young's inequality, it follows that if $\nu_\lambda \in L^p$, then $\nu_{\lambda^{1/m}}$ has a continuous density for an integer m satisfying $m \geq p/(p-1)$. It is therefore a direct consequence of Theorem 1.1 that if ν_λ is absolutely continuous, then $\nu_{\lambda^{1/m}}$ has a continuous density for sufficiently large m. As we are moreover able to make p quantitative, we are led to fully explicit examples of Bernoulli convolutions with a continuous density.

Denote by M_λ the Mahler measure of an algebraic number λ. Garsia proved in 1962 that ν_λ has an L^∞ density if $\lambda \in (1/2, 1)$ is a Garsia number, that is, λ satisfies $M_\lambda = 2$. There was no progress on understanding absolute continuity for explicit λ until Varjú [Var19] proved in 2019 that ν_λ is absolutely continuous if $\lambda > 1 - c \min\{\log M_\lambda, (\log M_\lambda)^{-2}\}$ for $c > 0$ an absolute constant. Varjú's result was improved in [KK25], addressing general self-similar measures, by showing that ν_λ is absolutely continuous if $\lambda > 1 - c \min\{\log M_\lambda, (\log \log M_\lambda)^{-2}\}$ for $c > 0$ an absolute constant. Yu [Yu22] proved in 2022 that ν_λ has a continuous density if λ is a Garsia number satisfying $\lambda \in [1/\sqrt{2}, 1)$.

In this paper, we give the first explicit non-Garsia examples of Bernoulli convolutions with a continuous density.

Theorem 1.2. *If $\lambda \in (1/2, 1)$ is algebraic of degree d_λ, has Mahler measure M_λ and satisfies*

$$\lambda \geq 1 - \left(\frac{\min\{1, \log M_\lambda\}}{1 + d_\lambda + \log M_\lambda} \right)^{100}, \tag{1.2}$$

then ν_λ is absolutely continuous with a density in $C_c(\mathbb{R})$.

We further discuss the proof of Theorem 1.2 in Section 3 and note that the exponent 100 is chosen to make the argument elegant. In fact, a refinement available at [KK] of the methods presented in this paper allow to lower the exponent to 3.

Theorem 1.3. *([KK]) If $\lambda \in (1/2, 1)$ is algebraic of degree d_λ, has Mahler measure M_λ and satisfies*

$$\lambda \geq 1 - \left(\frac{\min\{1, \log M_\lambda\}}{1 + d_\lambda + \log M_\lambda} \right)^3, \tag{1.3}$$

then ν_{λ^2} is absolutely continuous with a density in $L^2(\mathbb{R})$ and therefore ν_λ has a density in $C_c(\mathbb{R})$.

When one applies Theorem 1.3 to rational numbers, one obtains the following.

Corollary 1.4. *If $\lambda = 1 - p/q \in (1/2, 1)$ is in lowest terms and*

$$p \leq \frac{q}{(2 + \log q)^3},$$

then ν_λ is absolutely continuous with a density in $C_c(\mathbb{R})$. In particular, $\nu_{1-\frac{1}{n}}$ is absolutely continuous with a density in $C_c(\mathbb{R})$ for all $n \geq 588$.

Proof. As $d_\lambda = 1$ and $M_\lambda = q \geq 3$ the claimed inequality follows from Theorem 1.3. The desired result for $\nu_{1-\frac{1}{n}}$ follows from a brief calculation. □

Machine Contribution Statement. The proofs of this paper are due to GPT-6 Astra. On September 10th 2026, the second-named author (C.K.) asked GPT-6 Astra to find explicit examples of self-similar measures with an L^p density resulting in the discovery of Theorem 1.1. Various simplifications led to Lemma 2.1. Realising that Theorem 1.1 is quantitative, numerous conversations led to Theorem 1.2. The authors rewrote the paper and take responsibility for its correctness. A talk by the second-named author on the results of this paper is available at [Kog26].

Lean Formalization. GPT-6 Astra formalized all results from this paper and all necessary results from previous work. The Lean source code is available in the accompanying repository (todo).

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2. Proof of Theorem 1.1

Write $|E|$ for the Lebesgue measure of a subset $E \subset \mathbb{R}^d$. All balls considered are open. If $B = B(x, r)$ is an open ball around $x \in \mathbb{R}^d$ and $r > 0$, then we denote for $a > 0$ by

$$aB = B(x, ar).$$

The main idea of the paper is to apply the following lemma to self-similar measures. It is a form of the weak A_∞ criterion of Spadaro [Spa12, Theorem 1.1], [KKM22, Theorem 3.1 and Section 4] or [AHT17, Corollary 5.5].

Lemma 2.1 (Weak reverse Hölder criterion)**.** *For every $d \geq 1$ write $C = 18^d$. Let $f \geq 0$ belong to $L^1(\mathbb{R}^d)$. Suppose that for some $\varepsilon \in (0, 1)$ and $\eta \in (0, C^{-1})$ it holds that*

$$|E| \leq \varepsilon |B| \quad \Longrightarrow \quad \int_E f \, dx \leq \eta \int_{2B} f \, dx \tag{2.1}$$

for every ball B of radius less than one and every measurable $E \subset B$. Then $f \in L^p(\mathbb{R}^d)$ for $p = 1 + \frac{\log(C\eta)^{-1}}{\log(C\varepsilon^{-1})} > 1$.

Proof of Lemma 2.1. Write

$$A = C\varepsilon^{-1}, \qquad \gamma = 2 \cdot 6^d \eta \quad \text{and} \quad F(t) = \int_{\{f>t\}} f\,dx.$$

We claim that, for all sufficiently large t,

$$F(At) \le \gamma F(t). \tag{2.2}$$

Let $E = \{f > At\}$. By the Lebesgue differentiation theorem, almost all $x \in E$ satisfy that $\lim_{r\to 0} \frac{1}{|B(x,r)|}\int_{B(x,r)} f\,dx = f(x) > At > 2t$. For each such point $x \in E$, choose the ball B_x centered at x with largest radius whose average of f is at least $2t$. Continuity of ball averages gives

$$\frac{1}{|B_x|}\int_{B_x} f\,dx = 2t \quad \text{and} \quad \frac{1}{|aB_x|}\int_{aB_x} f\,dx \le 2t \quad \text{where} \quad a > 1.$$

For sufficiently large t, every ball $3B_x$ has radius less than one, since $2t|B_x| \le \|f\|_1$.

Cover a compact subset $K \subset E$ of these Lebesgue points by finitely many such balls and then choose a selection of disjoint balls B_i with $K \subset \bigcup_i 3B_i$. Since for each i,

$$|E \cap 3B_i| \le \frac{1}{At}\int_{3B_i} f\,dx \le \frac{2}{A}|3B_i| \le \varepsilon|3B_i|,$$

the hypothesis (2.1) yields

$$\int_K f\,dx \le \eta \sum_i \int_{6B_i} f\,dx \le 2 \cdot 6^d \eta\, t \sum_i |B_i| = \gamma t \sum_i |B_i|.$$

On the other hand, $2t|B_i| = \int_{B_i} f\,dx \le t|B_i| + \int_{B_i\cap\{f>t\}} f\,dx$. Disjointness therefore gives $t\sum_i |B_i| \le F(t)$. By inner regularity, $F(At) \le \gamma F(t)$, proving (2.2).

Choose $T > 0$ so that (2.2) holds for $t \ge T$, and write $I_R = \int_T^R t^{p-2}F(t)\,dt$. Note that $\gamma A^{p-1} = \frac{2\cdot 6^d\eta}{C\eta} = 2\left(\frac{1}{3}\right)^d \le \frac{2}{3}$. Therefore, for $R \ge AT$, changing variables in (2.2) gives $I_R \le I_{AT} + \gamma A^{p-1} I_{R/A} \le I_{AT} + \frac{2}{3} I_R$. Thus $I_R \le 3I_{AT}$ and hence I_R is uniformly bounded. Letting $R \to \infty$ and using $F \le \|f\|_1$, Tonelli's theorem gives

$$\frac{1}{p-1}\int_{\mathbb{R}^d} f^p\,dx = \int_{\mathbb{R}^d} f(x)\left(\int_0^{f(x)} t^{p-2}\,dt\right)dx = \int_0^\infty t^{p-2}F(t)\,dt < \infty.$$

This proves that $f \in L^p(\mathbb{R}^d)$ and so concludes the proof. □

Theorem 1.1 follows by applying Lemma 2.1 together with self-similarity.

Proof of Theorem 1.1. Without loss of generality we may normalise ν such that $\operatorname{diam} K = 1$, where $K = \operatorname{supp}\nu$. For a similarity g, we write $\rho(g)$ for the contraction factor of g and denote

$$\rho_* = \min_i \rho(g_i) > 0 \quad \text{and} \quad \omega(t) = \sup\{\nu(E) : |E| \le t\}.$$

Absolute continuity gives $\omega(t) \to 0$ as $t \downarrow 0$. We claim that for every ball $B = B(x,r)$ with $0 < r < 1$ and every measurable $E \subset B$,

$$\nu(E) \le \omega\left(c\frac{|E|}{|B|}\right)\nu(2B) \qquad \text{for} \qquad c = |B(0,1)|\rho_*^{-d}. \tag{2.3}$$

Since $\omega(t) \to 0$, by (2.3) it follows that the density function of ν verifies the hypothesis of Lemma 2.1 for every $\eta > 0$, which proves the theorem.

So it remains to prove (2.3). Let $\gamma_1, \gamma_2, \cdots$ be independent samples from μ. Write $q_n = \gamma_1 \cdots \gamma_n$ and consider the stopping time $\tau_r = \min\{n \geq 1 : \rho(q_n) \leq r\}$ and note $\rho_* r < \rho(q_{\tau_r}) \leq r$. It is well known that $q_{\tau_r}\nu$ has the same distribution as ν, as is shown in [Kit25, Lemma 2.19], in other words $\mathbb{E}\big[(q_{\tau_r})_*\nu\big] = \nu$.

Consider a set $E \subset B$. Observe that $|q_{\tau_r}^{-1}E| = \rho(q_{\tau_r})^{-d}|E| \leq c\frac{|E|}{|B|}$. Moreover, for $\nu(q_{\tau_r}^{-1}E) \neq 0$ it must hold that $q_{\tau_r}K \cap B \neq \emptyset$. So consider the event $\mathcal{H} = \{q_{\tau_r}K \cap B \neq \varnothing\}$. On $\mathcal{H}$, the set $q_{\tau_r}K$ has diameter at most r, so it is contained in $2B$ and therefore $\mathbb{P}(\mathcal{H}) \leq \mathbb{E}\big[\nu(q_{\tau_r}^{-1}(2B))\big] = \nu(2B)$. Thus, it follows by the definition of ω,

$$\nu(E) = \mathbb{E}\big[\nu(q_{\tau_r}^{-1}E)\big] \leq \omega\bigg(c\frac{|E|}{|B|}\bigg)\,\mathbb{P}(\mathcal{H}) \leq \omega\bigg(c\frac{|E|}{|B|}\bigg)\,\nu(2B),$$

which proves (2.3). □

3. Bernoulli convolutions

Throughout this section, write d_λ for the degree of an algebraic number $\lambda \in (0,1)$ and

$$\chi_\lambda = -\log\lambda \qquad \text{and} \qquad B_\lambda = \frac{1 + d_\lambda + \log M_\lambda}{\min\{1, \log M_\lambda\}}.$$

Thus (1.2) is equivalent to $(1-\lambda)B_\lambda^{100} \leq 1$. All logarithms are natural. The main result of this section is the following proposition.

Proposition 3.1 (A quantitative gain over L^1)**.** *If $\lambda \in (1/2, 1)$ is algebraic and $\chi_\lambda B_\lambda^{10} \leq 10^{-100}$, then ν_λ is absolutely continuous with density f_λ and*

$$f_\lambda \in L^{1+\delta_\lambda}(\mathbb{R}) \quad \text{for} \quad \delta_\lambda = \frac{1}{10(1 - \log(1-\lambda))}. \tag{3.1}$$

In Section 3.1 we deduce Theorem 1.2 from Proposition 3.1 by applying Young's inequality and then prove Proposition 3.1 in Section 3.2 and Section 3.3. The proof of Proposition 3.1 uses the detail of a measure, which is a concept introduced by the first-named author [Kit24], and ideas from Varjú [Var19] and the authors' paper [KK25]. As the proof of Proposition 3.1 is an illuminating series of ideas, we will not further outline it and simply let it unfold as presented below.

3.1. Deduction of Theorem 1.2.

We will use Young's inequality in the following form.

Lemma 3.2 (Young's inequality for scaled convolutions)**.** *Let $k \geq 2$ be an integer, let $a_1, \ldots, a_k > 0$, and let ν be a compactly supported absolutely continuous probability measure on $\mathbb{R}$ with a density in $L^p(\mathbb{R})$ for some $p \geq k/(k-1)$. Then*

$$\mathop{*}_{j=1}^{k} (D_{a_j})_*\nu$$

has a density in $C_c(\mathbb{R})$, where $D_a(x) = ax$.

Proof. For $1 \leq j \leq k$, let f_j be the density of $(D_{a_j})_*\nu$. Compact support and $p \geq k/(k-1)$ imply $f_j \in L^{k/(k-1)}$. By iterating Young's convolution inequality, the function $g = f_1 * \cdots * f_{k-1}$ is in L^k since $\frac{k-1}{k/(k-1)} - (k-2) = \frac{1}{k}$. Write $h = g * f_k$. Hölder's inequality gives $\|h\|_\infty \leq \|g\|_k \|f_k\|_{k/(k-1)}$ and, for every $x, t \in \mathbb{R}$,

$$|h(x+t) - h(x)| \leq \|g\|_k \, \|f_k(\cdot + t) - f_k\|_{k/(k-1)}.$$

The right-hand side tends to zero as $t \to 0$, by continuity of translations in $L^{k/(k-1)}$. Hence h is uniformly continuous. Moreover, its support is contained in the sum of the compact supports of the factors, so h is compactly supported. □

We will now deduce Theorem 1.2 from Proposition 3.1 and Lemma 3.2.

Proof of Theorem 1.2. We let $\theta = \lambda^k$ for some positive integer k we will choose later. We wish to show that θ satisfies the conditions of Proposition 3.1 and for Lemma 3.2 that

$$\delta_\theta \geq \frac{1}{k-1}. \tag{3.2}$$

First note that since $\theta \leq \lambda$ we have $\delta_\theta \geq \delta_\lambda$. Therefore for (3.2) to hold it is sufficient that $k \geq \frac{1}{\delta_\lambda} + 1 = 11 - 10\log(1-\lambda)$. Thus, we take

$$k = \lfloor 20 - 20\log(1-\lambda) \rfloor$$

and claim that

$$\chi_\theta B_\theta^{10} \leq 10^{-100}. \tag{3.3}$$

As $B_\theta \geq 1$, (3.3) implies $\theta \in (1/2, 1)$ and so Proposition 3.1 gives $f_\theta \in L^{1+\delta_\theta}$. Splitting the defining series modulo k gives $\nu_\lambda = *_{j=0}^{k-1}(D_{\lambda^j})_*\nu_\theta$, so (3.2) and Lemma 3.2 give a continuous density for ν_λ, proving the theorem.

To prove (3.3), note that standard height identities [BG06, Section 1.5 and Section 1.6] give

$$\frac{\log M_\theta}{d_\theta} = k\frac{\log M_\lambda}{d_\lambda} \quad \text{and} \quad \frac{d_\lambda}{k} \leq d_\theta \leq d_\lambda.$$

Hence $\log M_\lambda \leq \log M_\theta \leq k \log M_\lambda$ and $B_\theta \leq kB_\lambda$. Write $t = -\log(1-\lambda)$. Assumption (1.2) gives $e^{-t} = 1 - \lambda \leq B_\lambda^{-100}$ and therefore

$$t \geq 100 \log B_\lambda \quad \text{and} \quad B_\lambda^{10} \leq e^{t/10}.$$

The general estimate Lemma 3.10 below also gives $t \leq B_\lambda$. Since $B_\lambda \geq 3$, we have $t > 100$, hence $B_\lambda > 100$ and $t > 100 \log 100 > 400$. Using $k \leq 20(1+t)$ and $\chi_\lambda \leq 2(1-\lambda) = 2e^{-t}$, we obtain

$$\begin{aligned}
\chi_\theta B_\theta^{10} &\leq k^{11}\chi_\lambda B_\lambda^{10} \\
&\leq 2 \cdot 20^{11}(1+t)^{11}e^{-9t/10} \\
&\leq 2 \cdot 20^{11} \cdot 401^{11} e^{-360} < 10^{-100},
\end{aligned}$$

where we used that $(1+t)^{11}e^{-9t/10}$ is decreasing for $t \geq 400$. This concludes the proof. □

3.2. **The reduction to bounded approximation.** We now show how Lemma 2.1 is used. Indeed, Lemma 2.1 reduces Proposition 3.1 to finding a density function satisfying the property of the following lemma.

Lemma 3.3. *Let ν_λ be absolutely continuous with density function f_λ. Assume that there is a function g_λ satisfying*

$$\|f_\lambda - g_\lambda\|_1 \leq \frac{1}{2 \cdot 21^2} \quad \textit{and} \quad \|g_\lambda\|_\infty \leq \tfrac{1}{2}. \tag{3.4}$$

Then

$$f_\lambda \in L^{1+\delta_\lambda}(\mathbb{R}) \quad \textit{for} \quad \delta_\lambda = \frac{1}{10(1-\log(1-\lambda))}. \tag{3.5}$$

Proof. Normalize the support of ν_λ to $I = [0,1]$ and denote the resulting measure and functions by $\sigma = F_\lambda\,dx$ and G_λ. Since the original support has length $2/(1-\lambda)$, we have

$$\|F_\lambda - G_\lambda\|_1 \le \frac{1}{2\cdot 21^2} \quad \text{and} \quad \|G_\lambda\|_\infty \le K := (1-\lambda)^{-1}.$$

Thus $\omega(t) := \sup_{|E|\le t}\sigma(E) \le (2\cdot 21^2)^{-1} + Kt$. By (2.3), applied to σ with $c = 2/\lambda$, the hypothesis of Lemma 2.1 holds with

$$\eta = 21^{-2} \quad \text{and} \quad \varepsilon = \frac{\lambda}{4\cdot 21^2 K}.$$

Consequently, $F_\lambda \in L^p$ with

$$p - 1 = \frac{\log 21}{2\log(4\cdot 21^3 K/\lambda)} \ge \frac{1}{10(1+\log K)} = \delta_\lambda,$$

where we used $\lambda > 1/2$ and $K \ge 1$. Compact support and affine rescaling give (3.1). □

Next we recall the definitions of detail and higher order detail from [Kit24, Kit25]. We will use detail to establish the assumptions of Lemma 3.3 and then deduce Proposition 3.1. Let η_y be the centred Gaussian density of variance y. We will write $g_\lambda = \nu_\lambda * \eta_1$, whose supremum is at most $(2\pi)^{-1/2} < 1/2$. Write $\eta_y^{(k)} = \partial_y^k \eta_y$ and set

$$Q = \|\eta_1^{(1)}\|_1^{-1} = \sqrt{\pi e/2} \in (2, 21/10).$$

For a probability measure σ on $\mathbb{R}$, $r > 0$ and an integer $k \ge 1$, define

$$s_r^{(k)}(\sigma) = r^{2k} Q^k \|\sigma * \eta_{kr^2}^{(k)}\|_1 \quad \text{and} \quad s_r(\sigma) = s_r^{(1)}(\sigma).$$

We refer the reader to [Kit24, Kit25] and [KK25] for a discussion of detail and its properties. We just highlight that for $0 < a < b$,

$$\|\sigma * \eta_{a^2} - \sigma * \eta_{b^2}\|_1 \le \frac{2}{Q}\int_a^b s_t(\sigma)\,\frac{dt}{t}. \tag{3.6}$$

We first establish the following inequality, which is a more efficient version of [Kit25, Lemma 1.14] and of [KK25, Proposition 4.5].

Lemma 3.4. *Suppose that σ is a probability measure on $\mathbb{R}$, that k is a positive integer and that $r, b > 0$ with $b \ge r/\sqrt{k}$. Then*

$$s_r(\sigma) \le 10^k \left(\sup_{r/\sqrt{k}\le t\le b} s_t^{(k)}(\sigma) + \frac{r^2}{b^2} \right).$$

Proof. The case $k = 1$ is immediate so assume $k \ge 2$. Repeated integration by parts gives

$$\begin{aligned}
\frac{(-1)^{k-1}}{(k-2)!}\int_{r^2}^\infty (y-r^2)^{k-2}\sigma * \eta_y^{(k)}\,dy &= \frac{(-1)^{k-2}}{(k-3)!}\int_{r^2}^\infty (y-r^2)^{k-3}\sigma * \eta_y^{(k-1)}\,dy \\
&= \dots \\
&= -\int_{r^2}^\infty \sigma * \eta_y^{(2)}\,dy \\
&= \sigma * \eta_{r^2}^{(1)}.
\end{aligned}$$

Since $\frac{d}{dy}(1-\frac{r^2}{y})^{k-1} = (k-1)r^2(y-r^2)^{k-2}y^{-k}$, the kernel $(k-1)r^2(y-r^2)^{k-2}y^{-k}\,dy$ is a probability measure on $[r^2,\infty)$. For $b \geq r/\sqrt{k}$, its mass above kb^2 is, by Bernoulli's inequality,

$$1-\left(1-\frac{r^2}{kb^2}\right)^{k-1} \leq \frac{k-1}{k}\frac{r^2}{b^2} \leq \frac{r^2}{b^2}.$$

Since $Q^{1-k}k^k/(k-1)! < 10^k$, splitting at kb^2 and using that $s_r^{(k)}(\sigma) \leq 1$ yields

$$s_r(\sigma) \leq 10^k \left(\sup_{r/\sqrt{k}\leq t\leq b} s_t^{(k)}(\sigma) + \frac{r^2}{b^2}\right),$$

concluding the proof. □

Towards applying Lemma 3.3, we note that the following estimates of detail are sufficient.

Lemma 3.5 (Integrating detail)**.** *Let $R_0 \geq 100$ and $\delta > 0$. Suppose that a probability measure σ satisfies for $R \geq R_0,\ 1 \leq k \leq 10\log R$ and $r \in (0,1)$ that*

$$s_{e^{-R}}^{(k)}(\sigma) \leq 100^{-k} \quad \text{and} \quad s_r(\sigma) \leq \delta. \tag{3.7}$$

Then σ is absolutely continuous with density f and

$$\|f - \sigma * \eta_1\|_1 \leq 4\delta R_0 + \frac{1}{4R_0}. \tag{3.8}$$

Proof. We will let $R \geq 4R_0$ and apply Lemma 3.4 with $r = e^{-R}$, $k = \lceil 2\log R\rceil$ and $b = e^{-R/2}$. For $t \in [r/\sqrt{k}, b]$, we have $\log(1/t) \geq R/2 \geq R_0$ and $k \leq 10\log(R/2)$. Thus (3.7) applies uniformly on this interval, bounding the supremum by 100^{-k}. Therefore, for $R \geq 4R_0$,

$$s_{e^{-R}}(\sigma) \leq 10^k(100^{-k} + e^{-R}) \leq R^{-2}. \tag{3.9}$$

By [Kit24, Lemma 1.18], σ is absolutely continuous and denote by $f \in L^1(\mathbb{R})$ the density of σ. Since $\sigma * \eta_{a^2} \to f$ in L^1 as $a \to 0$ and letting $a \to 0$ in (3.6) with $b = 1$, gives

$$\begin{aligned}\|f - \sigma * \eta_1\|_1 &\leq \frac{2}{Q}\int_0^1 s_t(\sigma)\,\frac{dt}{t} \\ &\leq \frac{2}{Q}\left(\int_0^{4R_0} \delta\,dR + \int_{4R_0}^\infty \frac{dR}{R^2}\right) \leq 4\delta R_0 + \frac{1}{4R_0},\end{aligned}$$

where we substituted $t = e^{-R}$, used $2/Q < 1$ and the inequality (3.9) and the assumption. This proves (3.8). □

In the next subsection, we will establish the estimates necessary for Lemma 3.5 with suitable constants as stated in the following proposition.

Proposition 3.6 (Effective detail)**.** *If $\lambda \in (1/2,1)$ is algebraic and $\chi_\lambda B_\lambda^{10} \leq 10^{-100}$, then with $R_0 = 10^6 B_\lambda^3$ we have for $R \geq R_0$ and $1 \leq k \leq 10\log R$ that*

$$s_{e^{-R}}^{(k)}(\nu_\lambda) \leq 100^{-k}. \tag{3.10}$$

Moreover, for every $\lambda \in (1/2,1)$ and $r \in (0,1)$,

$$s_r(\nu_\lambda) \leq 40\chi_\lambda^{1/3}. \tag{3.11}$$

Having established all the above and in particular assuming Proposition 3.6, we prove Proposition 3.1.

Proof of Proposition 3.1. Set $R_0 = 10^6 B_\lambda^3$. Note $B_\lambda \geq 2 + d_\lambda \geq 3$ and hence $R_0 \geq 100$. Proposition 3.6 shows that ν_λ satisfies the hypotheses of Lemma 3.5 with this value of R_0 and $\delta = 40\chi_\lambda^{1/3}$. Thus ν_λ is absolutely continuous with density f_λ. Writing $g_\lambda = \nu_\lambda * \eta_1$, (3.8) gives

$$\begin{aligned}\|f_\lambda - g_\lambda\|_1 &\leq 160\chi_\lambda^{1/3} R_0 + \frac{1}{4R_0} \\ &\leq 160 \cdot 10^6 \cdot 10^{-100/3} + \frac{1}{108 \cdot 10^6} < \frac{1}{2 \cdot 21^2},\end{aligned}$$

where we used $B_\lambda \geq 3$ and $\chi_\lambda^{1/3} B_\lambda^3 = (\chi_\lambda B_\lambda^{10})^{1/3} B_\lambda^{-1/3} \leq 10^{-100/3}$ by the assumption of Proposition 3.1 that $\chi_\lambda B_\lambda^{10} \leq 10^{-100}$. Moreover, since ν_λ is a probability measure, $\|g_\lambda\|_\infty \leq \|\eta_1\|_\infty = (2\pi)^{-1/2} < \frac{1}{2}$. Thus f_λ and g_λ satisfy (3.4) and Lemma 3.3 gives $f_\lambda \in L^{1+\delta_\lambda}(\mathbb{R})$, as required. □

3.3. **Proof of Proposition 3.6: Effective detail from entropy and variance.** In this section we use ideas similar to [KK25] and also using

Lemma 3.7 (Small summands)**.** *Let $0 < \epsilon \leq 1$ and let $U_1, \ldots, U_n$ be independent real random variables satisfying $|U_i| \leq \epsilon r$ and write $v = \sum_i \mathrm{Var}(U_i)$. Then*

$$s_r\left(\sum_i U_i\right) \leq \frac{r^2}{r^2 + v} + 30\epsilon.$$

Proof. The proof is similar to the arguments in [KK25, Lemmas 4.7 and 4.10]. Put $v_i = \mathrm{Var}(U_i)$ and $y_i = r^2 + \sum_{j<i} v_j$. Replace the centred variables $U_i - \mathbb{E}[U_i]$ successively by independent centred Gaussians Z_i of variances v_i. The previous replacements smooth the kernel to $\eta_{y_i}^{(1)}$. The Hermite bound and the moment estimates give

$$\|\partial_x^3 \eta_y^{(1)}\|_1 \leq \frac{\sqrt{120}}{2} y^{-5/2}, \qquad \mathbb{E}[|U_i - \mathbb{E}[U_i]|^3] + \mathbb{E}[|Z_i|^3] \leq 4\epsilon r v_i.$$

Since $v_i \leq \epsilon^2 r^2$, comparison with an integral gives

$$\sum_i v_i y_i^{-5/2} \leq \frac{2}{3}(1 + \epsilon^2)^{5/2} r^{-3}.$$

Taylor expansion to order two, with matching moments, therefore bounds the error in detail by

$$\frac{2Q\sqrt{120}}{9}(1 + \epsilon^2)^{5/2}\epsilon < 30\epsilon.$$

Now apply the Gaussian identity in (**??**). □

The very small digits already provide weak smoothing at every scale, as expressed by (3.11). Indeed, retain a tail whose largest coefficient lies in $[\lambda r \chi_\lambda^{1/3}, r\chi_\lambda^{1/3}]$. Since $1 - \lambda^2 \leq 2\chi_\lambda$, its variance satisfies $v \geq r^2 \chi_\lambda^{-1/3}/8$. Lemma 3.7 bounds its detail by $(8 + 30)\chi_\lambda^{1/3}$. Convolve with the discarded digits; infinite tails follow by finite approximation.

Conditioning permits the same estimate when the summands themselves are large but lie in short conditional intervals.

Lemma 3.8 (Conditional variance)**.** *Put $\epsilon_0 = 10^{-4}$ and let $k \geq 1$ be an integer. Let $X_1, \dots, X_n$ have finite second moments and suppose that the σ-algebras $\sigma(X_i) \vee \mathcal{A}_i$ are independent. Assume that there are $\mathcal{A}_i$-measurable C_i such that*

$$|X_i - C_i| \leq \epsilon_0 r, \qquad \sum_i \mathbb{E}[\mathrm{Var}(X_i \mid \mathcal{A}_i)] \geq 1001 k r^2.$$

Then

$$s_r^{(k)}\left(\sum_i X_i\right) \leq 100^{-k}.$$

Proof. Put $W_i = \mathrm{Var}(X_i \mid \mathcal{A}_i)$. These variables are independent and lie in $[0, \epsilon_0^2 r^2]$. Greedy grouping supplies k disjoint groups with $m := \sum_i \mathbb{E}[W_i] \geq 1000 r^2$ in each group. For one such group, Chebyshev's inequality gives

$$\mathbb{P}\left(\sum_i W_i < m/2\right) \leq \frac{4\epsilon_0^2}{1000}.$$

Condition on all $\mathcal{A}_i$ and subtract the known centres. Lemma 3.7 and convexity bound the detail of the group by

$$\frac{2}{1000} + 30\epsilon_0 + \frac{4\epsilon_0^2}{1000} < \frac{1}{100}.$$

Apply (**??**) to the groups and convolve with the unused variables. □

For a finitely supported random variable X and $s > 0$, let

$$H(X; s) = \int_0^1 H(\lfloor X/s + t \rfloor)\, dt,$$

where H denotes Shannon entropy. If T is independent and uniform on $[0, 1]$, put

$$\mathcal{A}_s = \sigma(T, \lfloor X/s + T \rfloor), \qquad V(X; s) = s^{-2}\mathbb{E}[\mathrm{Var}(X \mid \mathcal{A}_s)].$$

The midpoint of the recorded interval is $\mathcal{A}_s$-measurable and lies within $s/2$ of X.

Lemma 3.9 (Entropy gives variance)**.** *For every finitely supported X and $s > 0$,*

$$H(X; s/4) - H(X; s) \leq 10 V(X; s)^{1/4}.$$

Proof. If Y is supported in an interval of length s, a uniformly shifted grid of mesh $s/4$ has at most five occupied cells. If Y' is an independent copy, the grid separates Y, Y' with probability $\min\{1, 4|Y - Y'|/s\}$. For a probability vector p on at most five points,

$$H(p) \leq 4\sqrt{1 - \sum_j p_j^2}.$$

Jensen's and Cauchy–Schwarz inequalities therefore give

$$H(Y; s/4) \leq 10(\mathrm{Var}(Y)/s^2)^{1/4}.$$

Now use independent uniform shifts T, U for the coarse and fine grids. Writing $A = \lfloor X/s + T \rfloor$ and $Z = \lfloor 4X/s + U \rfloor$, the chain rule gives

$$H(X; s/4) - H(X; s) = H(Z \mid T, U) - H(A \mid T, U) \leq H(Z \mid A, T, U).$$

Conditionally on (A, T), the variable X lies in an interval of length s and U remains independent. The preceding bound and Jensen's inequality finish the proof. □

Lemma 3.10 (Arithmetic input)**.** *Let $\lambda \in (1/2, 1)$ be algebraic, and write $X_\ell = \sum_{j=0}^{\ell-1} \varepsilon_j \lambda^j$. Then*

$$B_\lambda \geq 3, \qquad \Delta_\ell \geq e^{-B_\lambda \ell}, \qquad H(X_\ell) \geq \frac{\ell}{8B_\lambda}, \qquad \log \frac{1}{1-\lambda} \leq B_\lambda, \tag{3.12}$$

where Δ_ℓ is the minimum distance between distinct atoms.

Proof. Write $d = d_\lambda$, $M = M_\lambda$ and let $P_\lambda(T) = a \prod_{i=1}^{d} (T - \lambda_i)$ be the primitive integer minimal polynomial, with $a > 0$ and $\lambda_1 = \lambda$. Since

$$M \geq \frac{|P_\lambda(0)|}{\lambda} \geq \frac{1}{\lambda} > 1,$$

we have $3 \min\{1, \log M\} \leq 2 + \log M \leq 1 + d + \log M$, hence $B_\lambda \geq 3$. If P has coefficients in $\{-1, 0, 1\}$, degree $n < \ell$ and $P(\lambda) \neq 0$, its nonzero integer resultant gives

$$1 \leq a^n \prod_{i=1}^{d} |P(\lambda_i)|, \qquad |P(\lambda)| \geq (n+1)^{1-d} M_\lambda^{-n}.$$

Differences of digit sums are twice such values, so $\Delta_\ell \geq 2\ell^{1-d} M_\lambda^{-(\ell-1)} \geq e^{-B_\lambda \ell}$.

We prove the entropy bound directly for finite words. The case $\ell = 0$ is immediate, so suppose $\ell \geq 1$. If λ is not an algebraic unit, distinct sign words give distinct sums. Indeed, a collision gives a nonzero polynomial with coefficients in $\{-1, 0, 1\}$ vanishing at λ. Removing its lowest power of T and changing its sign if necessary gives a monic integer polynomial with constant coefficient ± 1, forcing λ to be an algebraic unit. Thus in the nonunit case $H(X_\ell) = \ell \log 2 \geq \ell/(8B_\lambda)$.

Now suppose that λ is an algebraic unit. Its minimal polynomial is monic, and the Mahler product formula gives

$$\log M = \sum_{i=1}^{d} \log \max\{1, |\lambda_i|\}.$$

Since

$$B_\lambda \log M = (1 + d + \log M) \frac{\log M}{\min\{1, \log M\}} \geq 1 + d + \log M > d,$$

some conjugate z satisfies

$$\log |z| \geq \frac{\log M}{d} > \frac{1}{B_\lambda}.$$

Take $k = \lceil B_\lambda \rceil$. Then $1 \leq k \leq 2B_\lambda$ and $|z^k| > e^{k/B_\lambda} \geq e > 2$.

Consequently, distinct sign words give distinct sums at parameter λ^k. To see this, a collision would yield a nonzero polynomial Q with coefficients in $\{-1, 0, 1\}$ and $Q(\lambda^k) = 0$. The minimal polynomial of λ then divides $Q(T^k)$ over $\mathbb{Q}$, so $Q(z^k) = 0$ as well. This is impossible: if $N = \deg Q$ and $|w| \geq 2$, the leading coefficient of Q has absolute value one, whereas

$$\left| \sum_{j=0}^{N-1} q_j w^j \right| \leq \sum_{j=0}^{N-1} |w|^j < |w|^N.$$

Choose a residue class modulo k containing the largest number m of indices in $\{0,\dots,\ell-1\}$. Then $m \ge \ell/k$. If its residue is s, write

$$Y = \sum_{j=0}^{m-1} \varepsilon_{s+jk}\lambda^{s+jk}, \qquad Z = X_\ell - Y.$$

The selected signs and the remaining signs are independent. Moreover, the injectivity just proved, together with $\lambda^s \neq 0$, shows that Y is uniform on 2^m distinct values. Hence

$$H(X_\ell) \ge H(X_\ell \mid Z) = H(Y) = m\log 2 \ge \frac{\ell\log 2}{k} \ge \frac{\ell\log 2}{2B_\lambda} \ge \frac{\ell}{8B_\lambda}.$$

This proves the entropy assertion of (3.12).

Finally, $1 \le |P_\lambda(1)| \le 2^{d-1}M_\lambda(1-\lambda)$, which gives the last bound in (3.12). □

The elementary argument above suffices because the complexity B_λ already contains the algebraic degree. Breuillard–Varjú [BV20] prove the stronger degree-independent entropy estimate in terms of Mahler measure alone; that stronger estimate is not needed here.

For rational $\lambda = a/b$, this input is elementary: the rational root theorem excludes coincidences between distinct digit words, giving $\Delta_\ell \ge 2b^{1-\ell}$ and $H(X_\ell) = \ell\log 2$. No entropy theorem is needed in that case.

Lemma 3.11 (Variance in a block)**.** *In the notation above, every integer $\ell \ge 100B_\lambda^2$ satisfies*

$$\int_{e^{-2B_\lambda\ell}}^{e^{-\ell/(32B_\lambda)}} V(X_\ell;s)\,\frac{ds}{s} \ge \frac{\ell}{10^{10}B_\lambda^7}. \tag{3.13}$$

Proof. Put $a = e^{-2B_\lambda\ell}$ and $b = e^{-\ell/(32B_\lambda)}$. At scales $t \le a$, the atoms are resolved, so $H(X_\ell;t) = H(X_\ell) \ge \ell/(8B_\lambda)$. Since $|X_\ell| \le (1-\lambda)^{-1}$, (3.12) gives, for $b/4 \le t \le b$,

$$H(X_\ell;t) \le \log\bigl(2(1-\lambda)^{-1}t^{-1}+2\bigr) \le B_\lambda + \frac{\ell}{32B_\lambda} + \log 10 \le \frac{\ell}{16B_\lambda}.$$

Telescoping yields

$$\int_a^b \bigl(H(X_\ell;s/4) - H(X_\ell;s)\bigr)\frac{ds}{s} = \int_{a/4}^{a} H(X_\ell;t)\frac{dt}{t} - \int_{b/4}^{b} H(X_\ell;t)\frac{dt}{t} \ge \frac{(\log 4)\ell}{16B_\lambda}.$$

The interval has logarithmic length at most $2B_\lambda\ell$. Lemma 3.9 and Hölder's inequality therefore give

$$\int_a^b V(X_\ell;s)\,\frac{ds}{s} \ge \frac{((\log 4)\ell/(16B_\lambda))^4}{10^4(2B_\lambda\ell)^3} > \frac{\ell}{10^{10}B_\lambda^7}. \qquad \square$$

Proof of Proposition 3.6. It remains to prove (3.10). Fix $r = e^{-R}$ with $R \ge R_0$. Let $\epsilon = 10^{-4}$ be as in Lemma 3.8, and put $N = \lceil 1000B_\lambda^2\rceil$. Use the lengths $\ell = N, N^2, \dots$ not exceeding $R/(4B_\lambda)$. We first place blocks of one length at the target scale, then separate the different lengths so that all blocks use disjoint digits.

One length. For a fixed ℓ, average grids of spacing $\ell\chi_\lambda$ in $v = \log(1/s)$ over the interval $[\ell/(32B_\lambda), 2B_\lambda\ell]$. Lemma 3.11 supplies a grid $\mathcal{S}_\ell$ with

$$\sum_{s\in\mathcal{S}_\ell} V(X_\ell;s) \ge \frac{1}{10^{10}\chi_\lambda B_\lambda^7}.$$

To move a mesh $s = e^{-v}$ on this grid to approximately ϵr, start a block of length ℓ at digit

$$n(s) = \left\lceil \frac{R + \log \epsilon^{-1} - v}{\chi_\lambda} \right\rceil, \qquad \lambda \epsilon r \le \lambda^{n(s)} s \le \epsilon r.$$

Here $n(s) \ge 0$, since $2B_\lambda \ell \le R/2$. Successive starting digits differ by ℓ, so these blocks are disjoint. Condition each using an independent shifted grid of mesh s. The scaled conditional radius is at most ϵr, and the total expected conditional variance contributed by this length is at least $\epsilon^2 r^2/(4 \cdot 10^{10} \chi_\lambda B_\lambda^7)$, since $\lambda^2 > 1/4$.

Several lengths. Since $\chi_\lambda \le 1/(64 B_\lambda)$, every digit j used by a block of length ℓ lies in the window

$$R + \log \epsilon^{-1} - j\chi_\lambda \in [\ell/(64B_\lambda),\, 2B_\lambda \ell].$$

These windows are disjoint for distinct lengths, since $N > 128 B_\lambda^2$. As $R/(4B_\lambda) \ge N$ and $\log(R/(4B_\lambda)) \ge \frac{1}{2} \log R$, the number of admissible lengths is at least

$$\left\lfloor \frac{\log(R/(4B_\lambda))}{\log N} \right\rfloor \ge \frac{\log R}{4 \log N}.$$

Adding the independent contributions and dividing by r^2 gives at least

$$\frac{\epsilon^2 \log R}{16 \cdot 10^{10} \chi_\lambda B_\lambda^7 \log N} > 10^5 \log R,$$

since $\log N \le B_\lambda^3$ and $\chi_\lambda B_\lambda^{10} \le 10^{-100}$. Lemma 3.8 applies for every $k \le 10 \log R$. Convolving with the unused digits proves (3.10). □

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